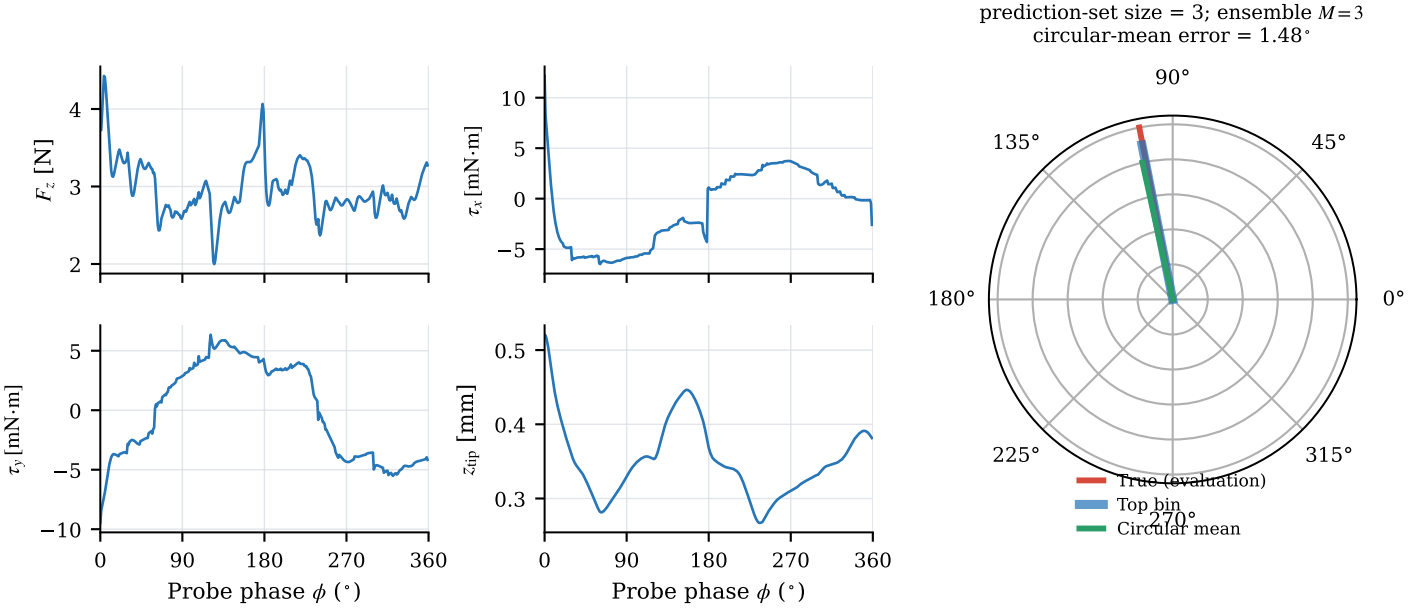

(a) Phase-indexed tactile fingerprint **(b) Circular direction summary**